Terms of Reference (ToR)
Pilot Design, Development, Deployment, and Evaluation of a Conversational AI-Based Early Warning System for Farmers Using the Agvisely Platform

Background

The International Maize and Wheat Improvement Center (CIMMYT) is a cutting-edge, non-profit international organization committed to addressing tomorrow's challenges today. As a core CGIAR Research Center, part of a global research partnership working toward a food-secure future, CIMMYT is dedicated to reducing poverty, enhancing food and nutrition security, and improving natural resource management.

Agvisely is a weather forecast–based agricultural advisory tool supporting crop, aquaculture, and livestock systems, and is a digital solution for climate-smart agriculture. In Bangladesh, where climate variability is high and farmers are frequently exposed to extreme weather events, strengthening resilience is critical. Agvisely addresses this challenge by delivering localized, timely weather forecasts and crop advisories to support effective climate adaptation at the farm level.

Developed through collaboration between CIMMYT, the Bangladesh Meteorological Department (BMD), and the Department of Agricultural Extension (DAE), Agvisely is a user-friendly Climate Information Service platform (mobile app) designed primarily for extension workers and rural farming communities.

Although the mobile application is well designed, tested, and validated, few user-level gaps remain. A large proportion of farmers use feature phones instead of smartphones, and many have limited or no access to mobile internet. In addition, low digital literacy further limits effective utilization of the platform. To address these challenges, there is a need to explore inclusive solutions such as AI-driven, voice-based (conversation) advisory systems in Bangla to ensure broader accessibility and impact.

CIMMYT is seeking qualified IT software development companies to design, develop, and integrate an AI-powered, voice-based Bangla conversational advisory system. This system should enable farmers to ask questions via mobile phones in Bangla and receive responses related to weather forecasts, agricultural advisories, and crop disease warnings strictly from the available information from Agvisely system, that generates location specific dynamic advisory. Also, the same system will access the wheat disease forecast based (pre-season) advisory, which is static information.

Objective:

The selected company will be responsible for achieving the following objectives:

1. Develop an AI-powered, voice-based advisory accessing Agvisely API that enables farmers to access weather forecasts, agricultural advisories (10 crops) and wheat disease foecasts through simple voice interactions in Bangla using basic mobile phones.
2. Ensure inclusive access to climate information service, particularly for farmers using

feature phones, with limited internet connectivity and low digital literacy.

3. Design and implement a comprehensive evaluation framework to assess the effectiveness of the AI advisory system, including user engagement, comprehension, trust, adoption of recommended practices, and agricultural outcomes, to support continuous improvement and evidence-based scaling.

Scope of work:
The selected company will be responsible for development, including coding, system design, and integration of a Bangla conversational AI (voice system) interface for farmer interaction linked to the Agvisely system. The system must integrate weather forecasts, agricultural advisories, and disease warnings.

The company will also evaluate system usability, accuracy, and impact on farmers' practices. Following validation, the AI-driven voice-based advisory system will be integrated into the Agvisely platform.

Design and development

1. System Design

Assess user needs (farmers, extension agents, and stakeholders). Define system architecture integrating:

- Functional Bangla conversational AI (voice system) that uses Agvisely Crop advisory; Crop disease warning
- Registering hotline numbers to access Agvisely AI (Conversation agent)

2. Development & Integration

- Develop conversational AI (voice system) modules.
- Integrate with Agvisely backend systems and APIs.
- Incorporate crop and early disease warning systems.
- Ensure compatibility with feature mobile devices and low-bandwidth environments.

Deliverables
The selected company shall deliver:

- Functional Bangla conversational AI (voice system) prototype.
- Creating a hotline number to access the system.
- Delivering advisories and diseases early warning to at least 1000 farmers and extension officers.
- Collected information on user engagement, comprehension and an anutomated survey on , trust, adoption of recommended practices, and agricultural outcomes.
- System architecture and design documentation.
- Final project report.

Accountability
After delivery, the company will fix any coding bugs related to the conversational AI (voice)

system, unless all source code is handed over.

Timeline

- 15 July – 15 August 2026: System design, development, and integration with Agvisely.
- 15 August – 30 November 2026: Pilot deployment, monitoring, and collection of evaluation data.